

Explicit Airfoil Mappings

Joukowski, Kármán–Trefftz, and Tsien

Complex Analysis

Motivation

We construct airfoil shapes via conformal mappings.

General idea:

- ▶ start with flow around a circle
- ▶ map the exterior of the circle
- ▶ obtain an airfoil geometry

We emphasize **explicit formulas with parameters**.

Joukowski Transformation

The Joukowski map is

$$w = z + \frac{c^2}{z}, \quad c > 0.$$

Domain:

- ▶ exterior of the circle $|z| = c$

Derivative:

$$\frac{dw}{dz} = 1 - \frac{c^2}{z^2}.$$

Critical points:

$$z = \pm c.$$

Trailing Edge and Geometry

At

$$z = c,$$

we have

$$\frac{dw}{dz} = 0.$$

Thus:

- ▶ the image has a **cusp**
- ▶ this is the trailing edge

Image point:

$$w = c + \frac{c^2}{c} = 2c.$$

Generating an Airfoil

Use an offset circle:

$$z(\theta) = z_0 + ce^{i\theta}, \quad 0 \leq \theta < 2\pi.$$

Then

$$w(\theta) = z_0 + ce^{i\theta} + \frac{c^2}{z_0 + ce^{i\theta}}.$$

Parameters:

- ▶ c controls size
- ▶ z_0 controls camber and asymmetry

Limitation of Joukowski Map

The trailing edge is always a cusp:

$$\text{angle} = 0.$$

Real airfoils require a finite trailing angle.

We generalize the mapping.

Kármán–Trefftz Transformation

Define

$$w = c \frac{(z + c)^\lambda + (z - c)^\lambda}{(z + c)^\lambda - (z - c)^\lambda}, \quad \lambda > 1.$$

This is a conformal mapping of the exterior of a circle to an airfoil with finite trailing edge angle.

Parameters and Geometry

Parameters:

- ▶ c sets scale
- ▶ λ controls trailing edge angle

Trailing edge angle:

$$\theta = \pi \left(1 - \frac{1}{\lambda} \right).$$

Special case:

$$\lambda = 1 \quad \Rightarrow \quad \text{Joukowski map.}$$

Circle Mapping

As before, take

$$z(\theta) = z_0 + ce^{i\theta}.$$

Then

$$w(\theta) = c \frac{(z(\theta) + c)^\lambda + (z(\theta) - c)^\lambda}{(z(\theta) + c)^\lambda - (z(\theta) - c)^\lambda}.$$

This produces a wide range of airfoil shapes.

Tsien Symmetric Transformation (1943)

A useful symmetric mapping is

$$w = z + \frac{c^2}{z} + \frac{\alpha c^4}{z^3},$$

where

$$\alpha \in \mathbb{R}.$$

This preserves symmetry about the real axis.

Role of Parameters in Tsien Map

In

$$w = z + \frac{c^2}{z} + \frac{\alpha c^4}{z^3},$$

- ▶ c controls overall size
- ▶ α controls thickness distribution

Unlike Joukowski:

- ▶ leading edge is smoother
- ▶ thickness is more flexible

Symmetric Airfoil Generation

Using

$$z(\theta) = ce^{i\theta},$$

we obtain

$$w(\theta) = ce^{i\theta} + ce^{-i\theta} + \alpha ce^{-3i\theta}.$$

Thus the airfoil is symmetric about the real axis.

Comparison

- ▶ Joukowski:

$$w = z + \frac{c^2}{z}$$

- ▶ simple
- ▶ cusp

- ▶ Kármán–Trefftz:

power-law mapping

- ▶ finite trailing angle

- ▶ Tsien:

$$w = z + \frac{c^2}{z} + \frac{\alpha c^4}{z^3}$$

- ▶ symmetric
- ▶ flexible thickness

Summary

We constructed explicit conformal mappings:

- ▶ Joukowski transform

$$w = z + \frac{c^2}{z}$$

- ▶ Kármán–Trefftz transform

finite-angle generalization

- ▶ Tsien symmetric transform

$$w = z + \frac{c^2}{z} + \frac{\alpha c^4}{z^3}$$

These are central in historical airfoil theory and potential flow.

Summary

Conformal mappings allow us to:

- ▶ transform simple geometries into airfoils
- ▶ analyze fluid flow via complex potentials

Important mappings:

- ▶ $w = z + \frac{1}{z}$
- ▶ generalized power-law maps
- ▶ refined symmetric mappings